

Information Visualization

LaTeX, Data Preprocessing, Visualization Pipelines

Course Objectives

- Master LaTeX for academic documents
- Learn data preprocessing techniques
- Build effective visualization pipelines
- Embrace "Logic over Layout" mindset

LaTeX vs. Word Processing

Two Fundamentally Different Approaches to Document Creation

WORD (Imperative)

"Paint" your text directly:

- Select text → Click Bold
- Change color manually
- Set font size to 12pt

Problems:

- Format "drift" in long docs
- Manual cross-refs break
- Inconsistent styling

LaTeX (Declarative)

"Label" your content:

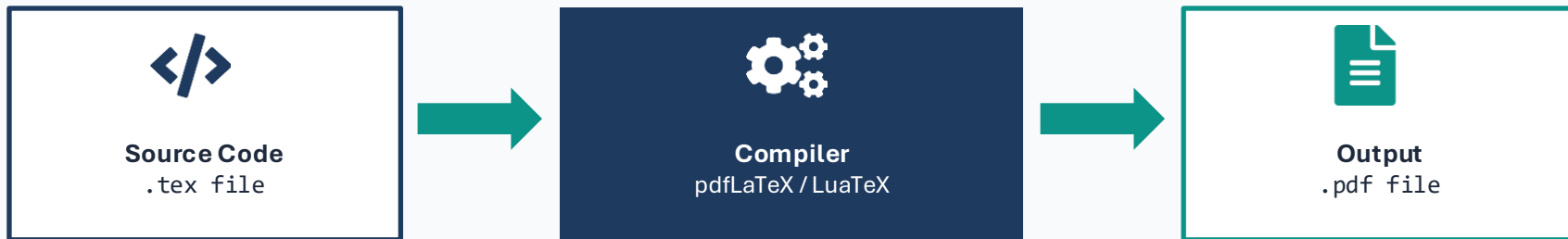
- `\section{Introduction}`
- `\emph{important}`
- `\ref{fig:results}`

Benefits:

- Consistent formatting
- Dynamic cross-references
- Perfect for 100+ pages

Key Insight: Define WHAT content IS, not HOW it looks. The engine handles formatting.

The Compilation Cycle



Key Difference from Real-Time Editing

Real-Time (Word/Docs)

- Changes visible immediately
- Errors accumulate silently
- Final output may have issues

LaTeX Compilation

- Errors stop the build
- Must fix errors to proceed
- Final PDF is perfect

Minimal Document Structure

```
\documentclass[12pt, a4paper]{article}

% --- Preamble (Load packages) ---
\usepackage[utf8]{inputenc}
\usepackage{amsmath}    % Math formulas
\usepackage{graphicx}   % Images

\title{My Academic Report}
\author{Data Scientist}
\date{\today}

\begin{document}
\maketitle
\section{Introduction}
This is a sample document.
\end{document}
```

1. Document Class

Type: article, report, book

2. Preamble

Packages & global settings before document begins

3. Document Body

All content between \begin and \end



This is the "Hello World" of LaTeX!

Essential LaTeX Packages

Packages extend LaTeX — like Python libraries

amsmath

Mathematical typesetting

- Equations & formulas
- Matrices & arrays
- Aligned equations

geometry

Page layout control

- Custom margins
- Paper size settings
- Landscape mode

hyperref

Hyperlinks & navigation

- Clickable TOC links
- URL hyperlinks
- PDF bookmarks

Usage Example:

```
\usepackage{amsmath}
\usepackage[margin=1in]{geometry}
\usepackage[colorlinks=true]{hyperref}
```

Understanding the Preamble

Everything before `\begin{document}` — the "brain" of your file

Global Settings

Document class, paper size, font size

Font Encoding

UTF-8 support, font packages

Package Configurations

Load and configure all packages

Custom Commands

Define reusable macros with `\newcommand`

Textual Commands

Lists

```
\begin{itemize}
  \item First point
  \item Second point
\end{itemize}
% Use enumerate for numbered
```

Emphasis & Formatting

```
\emph{italic text}
\textbf{bold text}
\texttt{monospace}
\underline{underlined}
```

Special Characters (Reserved)

Char	Code	Char	Code
%	\%	&	\&
\$	\\$	#	\#
—	_	{ }	\{ \}

Math Environments

Inline Math: $\$ \dots \$$

Flows with text:

The equation $x^2 + y^2 = r^2$ describes a circle.

Display Math: $\displaystyle \left[\dots \right]$

Centered on its own line:

$\displaystyle \left[E = mc^2 \right]$

Result: $E = mc^2$

Numbered Equations

```
\begin{equation}
  \label{eq:quad}
  x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}
\end{equation} % Reference: Equation~\ref{eq:quad}
```


The Algebra of Summations

$$\sum_{i=1}^n x_i = x_1 + x_2 + \dots + x_n$$

LaTeX Code:

```
\sum_{i=1}^n x_i$ % Inline: smaller sigma
```

Components:

\sum

Summation symbol (Σ)

_{i=1}^n

Limits (subscript/superscript)

x_i

Variable with index

Fractions and Roots

Fractions: `\frac{num}{den}`

$a/b \rightarrow \frac{a}{b}$

Complex: $\frac{x^2 + 1}{2x - 3}$

Roots: `\sqrt[n]{x}`

`\sqrt{x}` % Square root

`\sqrt[3]{x}` % Cube root

`\sqrt[n]{x}` % nth root

Nesting Commands

```
% A fraction inside a square root:  
\sqrt{\frac{a}{b}} % Result:  $\sqrt{a/b}$ 
```

Delimiters and Sizing

Standard parentheses look too small for tall fractions

✗ Without `\left \right`:
 (a/b) ← Brackets too small!

✓ With `\left \right`:
 (a/b) ← Brackets scale!

```
\left( \frac{a}{b} \right) % Brackets grow to match content
```

Available: `() [] {} | \langle \rangle`

Matrices and Arrays

Essential for datasets and linear transformations

pmatrix

() brackets

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

bmatrix

[] brackets

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

vmatrix

|/| brackets

$$\begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix}$$

Syntax: & = column separator, \\ = row break

```
\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}
```

Figures and Floating

LaTeX "floats" figures to best location — let the algorithm decide!

```
\begin{figure}[htbp]
  \centering
  \includegraphics[width=0.8\textwidth]
    {images/results.png}
  \caption{Results}
  \label{fig:results}
\end{figure}
```

Position Specifiers

h	Here
t	Top of page
b	Bottom
p	Float page

BibTeX Integration

Automated bibliography — never type references manually

The .bib File

```
@article{smith2023,  
  author={Smith, J.},  
  title={Data Analysis},  
  journal={J. Data Sci.},  
  year={2023}  
}
```

Usage & Tools

```
\cite{smith2023}  
→ Outputs: [1] or (Smith, 2023)
```

Tools that export .bib:

Zotero • Mendeley • EndNote

Professional Tables

Journals hate "grid" tables — use booktabs for clean look

 **Avoid**

Method	Acc
A	0.89
B	0.92

 **booktabs style**

Method	Acc
A	0.89
B	0.92

The booktabs Package

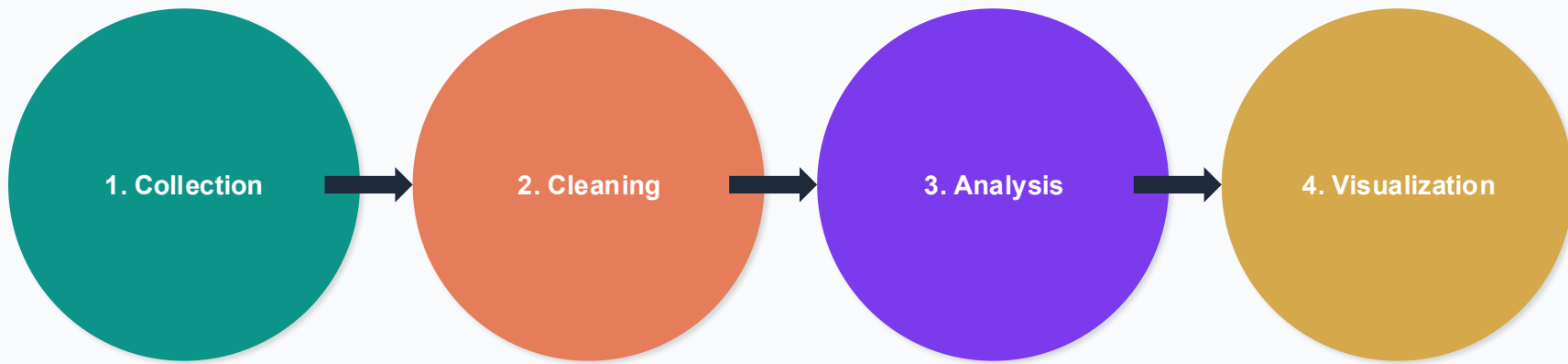
```
\toprule    % Thick top line
\midrule    % Thin separator
\bottomrule % Thick bottom line
```

Data Preprocessing & Cleaning

Transforming raw data into analysis-ready datasets

The Data Lifecycle

Cleaning is NON-LINEAR — you return after seeing visualization issues



The cycle is iterative — weird visualizations send you back to cleaning!

Dirty Data Symptoms

Missing Values

NULL, NaN, empty cells
Biased analysis

Inconsistent Labels

"USA" vs "United States"
Wrong groupings

Outliers

Extreme values
Skewed statistics

Example: Country Names

ID	Country
001	USA
002	United States
003	U.S.A.

Identifying Variable Scales

CATEGORICAL

Nominal

No order (Colors, Gender)

Ordinal

Has order (Ratings, Education)

NUMERICAL

Interval

Equal intervals (Temp °C)

Ratio

True zero (Height, Income)

ZIP codes are NUMBERS but NOMINAL — you can't take their mean!

Exploratory Data Analysis (EDA)

Find the "shape" of your data before cleaning

Summary Stats

Mean,
Median,
Std Dev
`df.describe()`

Distributions

Histograms,
Box plots
`sns.histplot()`

Correlations

Heatmaps,
Scatter
`df.corr()`

Missing Data: Deletion

Listwise: Remove row if ANY missing → `df.dropna()`

Pairwise: Use available data per calculation → `df.corr()`



Risk: Listwise can lose 50%+ of data if many columns have missing values!

Missing Data: Imputation

Simple: Mean/Mode

```
df['col'].fillna(df['col'].mean())
```

Predictive: K-NN

```
KNNImputer(n_neighbors=5)
```

- Domain-specific: Sometimes context provides the best fill
 - 0 for sales,
 - 'Not measured' for medical

Identifying Outliers

An outlier isn't always an error — it could be a discovery!

Standard Dev

$$|x - \mu| > 3\sigma$$

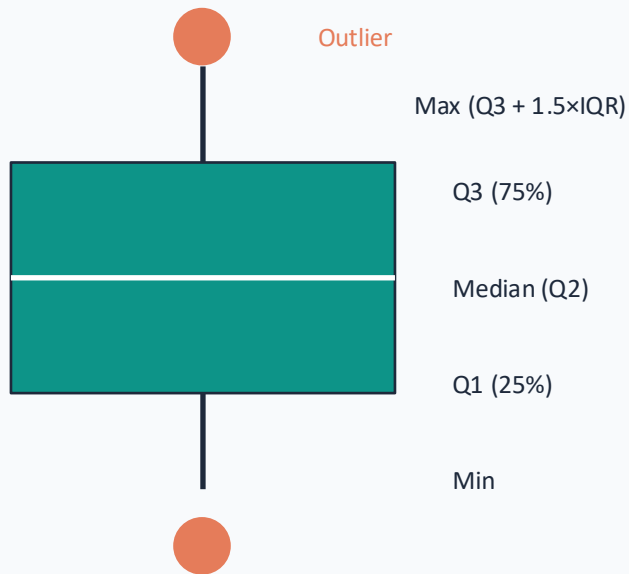
IQR Method

$$x > Q3 + 1.5 \times IQR$$

Z-Score

$$|Z| > 3$$

The Box Plot Anatomy



IQR (Interquartile Range) = $Q3 - Q1$ (The "box" = middle 50%)

Outlier if: $x < Q1 - 1.5 \times IQR$ OR $x > Q3 + 1.5 \times IQR$

The 1.5 multiplier is a statistical convention — catches ~0.7% of normal data

Why Scale Data?

Critical for: K-Means, KNN, Neural Networks, Gradient Descent

Problem: If Feature A ranges 0-1 and Feature B ranges 0-1,000,000

The model will IGNORE Feature A completely!

Min-Max Scaling

$$x' = (x - \min) / (\max - \min) \rightarrow \text{Range } [0, 1]$$

Standardization (Z-Score)

$$z = (x - \mu) / \sigma \rightarrow \text{Mean}=0, \text{Std}=1$$

Min-Max Scaling Code

```
import pandas as pd
from sklearn.preprocessing import MinMaxScaler

# Sample Data
data = {'Income': [45000, 54000, 80000, 120000]}
df = pd.DataFrame(data)

# Initialize and Transform
scaler = MinMaxScaler()
df['Income_Scaled'] = scaler.fit_transform(df[['Income']])
```

Result: 45000 → 0.0, 120000 → 1.0 (proportional between)

Standardization (Z-Score)

$$Z = (x - \mu) / \sigma$$

Result: Mean = 0, Standard Deviation = 1

Best for: Algorithms that assume Gaussian/Normal distributions (Linear Regression, SVM)

```
from sklearn.preprocessing import StandardScaler  
scaler = StandardScaler()
```

Handling Categorical Data

Label Encoding

Red → 0, Blue → 1, Green → 2

Creates false ordering!

One-Hot Encoding

Red: [1,0,0], Blue: [0,1,0]

✓ No false ordering for nominal data

Use One-Hot for NOMINAL data (no inherent order)

Date/Time Parsing

Time is often stored as STRING — convert to datetime for analysis

```
# Convert string to datetime
df['date'] = pd.to_datetime(df['date_str'])

# Extract components
df['year'] = df['date'].dt.year
df['month'] = df['date'].dt.month
df['day_of_week'] = df['date'].dt.dayofweek
```

ISO 8601 format: 2024-01-15T14:30:00Z (recommended standard)

Text Cleaning

Essential for NLP pipelines

Lowercasing

"Hello" → "hello"

Remove Punctuation

"Hi!" → "Hi"

Stop Words

Remove "the", "and", "is"

```
text.lower().translate(str.maketrans('', '', string.punctuation))
```

Data Integration (Joins)

When joining tables, ALWAYS check row count afterward!

Inner Join

Only matching rows from BOTH tables

Outer Join

All rows from BOTH (with NULLs)

```
result = pd.merge(df1, df2, on='key', how='inner')
```

Dealing with Skewness

Many real variables (wealth, population) follow power laws with long tails

Log Transformation

$$x' = \log(x)$$

Compresses large values

Power Transformation

Box-Cox, Yeo-Johnson
Optimizes for normality

```
df['income_log'] = np.log1p(df['income']) # log1p handles zeros
```


Data Quality Audits

Create a "Data Quality Report" for stakeholders

Completeness

% of non-missing values

Uniqueness

No duplicate records

Consistency

Same format across rows

Automation in Cleaning

NEVER clean data in Excel manually — use repeatable scripts!

- **Functional programming:** Pure functions, no side effects
- **Repeatable scripts:** Re-run when raw data changes
- **Pipeline objects:** `sklearn.pipeline.Pipeline`

Summary: Clean data → Tidy data → Ready for visualization

Visualization Pipelines

From data tables to meaningful visual insights

The Visualization Goal

Is the chart to help YOU find a trend, or help THEM understand a conclusion?

Discovery

Exploratory analysis

Finding patterns

For: Analyst/Researcher

Communication

Present findings

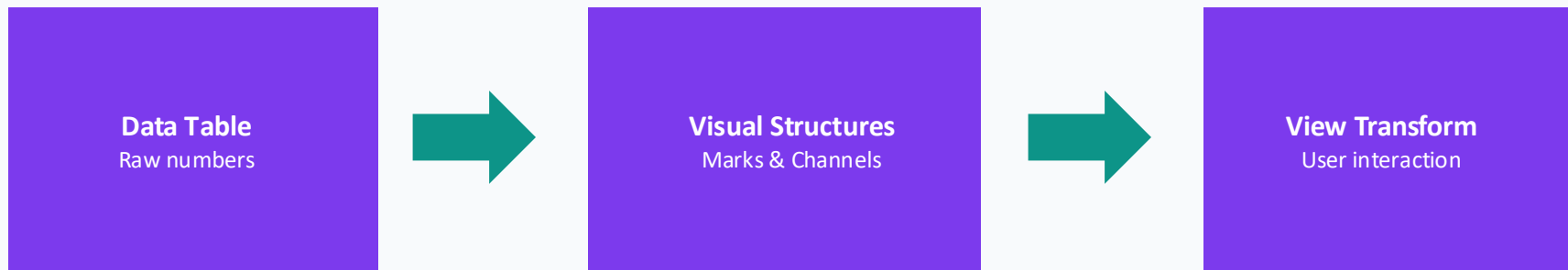
Tell a story

For: Stakeholders

Both require: Integrity + Perceptual Efficiency

Anatomy of a Pipeline

A systematic way to think: how does a NUMBER become a PIXEL?



Data Transformation

You rarely plot every row — you plot a REPRESENTATION

Filtering

Select subsets
`df[df['age'] > 30]`

Aggregating

Sum, Mean, Count
`df.groupby('cat').mean()`

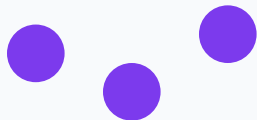
Binning

Create histograms
`pd.cut(df['val'], bins=10)`

Visual Mapping: Marks

40

Marks are the basic geometric elements on screen



Points
(Scatter plots)



Lines
(Trends, Time series)



Areas
(Bar charts, Maps)

Visual Mapping: Channels

Channels control HOW marks represent data values

Position

X/Y coordinates
Most accurate!

Color

Hue, Saturation
Good for categories

Size

Area, Length
Less precise

Shape

Circle, Square
Categories only

Human Perception Hierarchy

Humans judge lengths from common baseline BEST — why bar charts work

Accuracy Ranking (Best to Worst):

1. Position on common scale

2. Position on non-aligned scale

3. Length

4. Angle/Slope

5. Area

6. Color saturation

7. Color hue

Pie charts use ANGLES — one of the worst channels for accuracy!

Color Palettes in Pipelines

Choose palette type based on your data type

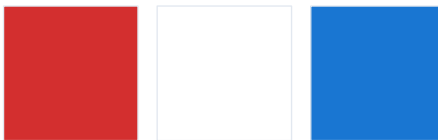
Sequential

Low → High
One hue intensifying



Diverging

Negative ← 0 → Positive
Two hues from midpoint



Qualitative

Categories
Distinct hues



✗ NEVER use rainbow palette for continuous data — creates false boundaries

Coordinate Systems

The choice of coordinates changes the entire "story" of your data

Cartesian (x, y)

Standard charts
Most common

Polar

Pie, Radar charts
Cyclical data

Geographic

Maps (Lat/Long)
Spatial patterns

The Grammar of Graphics

Framework (ggplot2): Treat a plot like a sentence with nouns and verbs

Data

The dataset

Aesthetics

x, y, color, size mappings

Geometries

Points, lines, bars

Scales

Axis ranges, color scales

Layers

Stack multiple geoms

Telling Stories with Data

A pipeline isn't finished until the "So what?" is clear

Annotations

Highlight key points directly on chart

Contextual Labels

Units, time periods, data sources

Narrative Flow

Guide viewer through the insight

The TITLE should answer the question, not just label the axes

High-Dimensional Visualization

How to show 10 variables on a 2D screen?

Scatterplot Matrix

Pairwise comparisons

$N \times N$ grid of plots

Parallel Coordinates

Each axis = one variable

Lines connect values

PCA (Dimensionality Reduction)

Find 2 most important
principal components

Interactive Pipelines

Modern pipelines let USERS filter the data themselves

Brushing & Linking

Select in one chart,
highlight in others

Hover Effects

Tooltips with details
on demand

Drill-downs

Click to see
more granular data

Tools: Plotly, D3.js, Tableau, Power BI, Dash

Design Ethics

Visualizations can LIE easily — remain honest

Don't truncate Y-axis

Starting at non-zero
exaggerates differences

Show uncertainty

Error bars, confidence
intervals

Accessibility

Color-blind friendly palettes

✓ Your job: Inform, not mislead